

Threat Fields: Additivity, Equivariance and the Pawn Anomaly

Theorem monograph T04 of the chess-dynamics-lab program —
the K3 layer

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Keywords: threat field, additivity, equivariance, anomaly

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Setting

The K3 layer of our three-layer particle model is the potential layer: every piece-particle emits a threat field on the board, and the full configuration defines the total field. The threat field is the function $\Theta_s(c)$ equal to the number of particles of side s attacking the square c . It is a discrete analogue of the potential created by a system of sources; its structure determines the defensive links, the pressure and the energy of a position.

The task of this monograph is to prove three fundamental properties of the field: additivity (the total mass of the field equals the sum of the particle mobilities — the attack identity), equivariance under the board group D_4 for orientation-neutral pieces, and the exact value of the pawn anomaly — the equivariance violation introduced by oriented particles (pawns attacking forward only). All three properties are verified by protocol C6.

Theorem

Theorem T04 (the structure of the threat field). (i) The attack identity: for any position p and side s , $\sum_c \Theta_s(c) = \sum_\pi a(\pi)$, the right sum over the particles π of side s , where $a(\pi)$ is the number of squares attacked by the particle π . (ii) For pawnless positions the threat field is D_4 -equivariant: $\Theta_{g \cdot p}(g \cdot c) = \Theta_p(c)$ for all $g \in D_4$ and all squares c . (iii) In the initial position the total pawn anomaly (the number of pairs (g, c) with $\Theta_{g \cdot p}(g \cdot c) \neq \Theta_p(c)$ over both sides) equals exactly 176.

Proof

(i) The attack identity is an interchange of the order of summation in the finite setting (a discrete Fubini theorem). The field $\Theta_s(c)$ equals by definition $\sum_{\pi}[c \in A(\pi)]$, where $A(\pi)$ is the set of squares attacked by the particle π and $[\cdot]$ is the indicator. Summing over c : $\sum_c \Theta_s(c) = \sum_c \sum_{\pi}[c \in A(\pi)] = \sum_{\pi} \sum_c [c \in A(\pi)] = \sum_{\pi} |A(\pi)| = \sum_{\pi} a(\pi)$. In the initial position each side carries the field mass 38: the exact count follows the blocking rule and is verified by the protocol. The identity guarantees the equivalence of the two counting modes: by squares and by particles.

(ii) Equivariance. The key observation: the attack relation is defined by the relative geometry of squares — belonging to one file/rank/diagonal with intermediate blocking, or a fixed offset for the knight, or adjacency for the king. All these relations are invariant under any isometric transformation of the board: g maps a line to a line, preserves betweenness (“lying between”) and maps the knight offset set to itself (the set $\{(\pm 1, \pm 2), (\pm 2, \pm 1)\}$ is invariant under all eight D_4 transformations — a direct check of eight cases). Formally, for a pawnless configuration $\Theta_{g \cdot p}(g \cdot c) = \sum_{\pi}[g \cdot c \in A(g \cdot \pi)] = \sum_{\pi}[c \in A(\pi)] = \Theta_p(c)$, where the middle equality is the invariance of the attack relation. Blocking transfers as well: a square lies between two others iff its image lies between the images.

(iii) The pawn anomaly. A white pawn attacks strictly~“forward” — the squares $(f \pm 1, r + 1)$; the direction is fixed by the side colour, not by the board geometry. No non-identity element of D_4 preserves the~“forward for White” direction: rotations and reflections carry the vertical direction into another one or flip its sign. Hence for positions with pawns the equivariance fails, and the magnitude of the failure is a finite, exactly computable number. In the initial position a direct enumeration of all $8 \times 64 = 512$ pairs (g, c) for each side yields 88 violations per side, in total $88 + 88 = 176$; the number is frozen in the baseline and reproduced by the protocol. The anomaly is non-zero but finite and structural: it shows that the pawn is an~“oriented particle” breaking the symmetry of the background. The theorem is proved.

Protocol data

The threat field (check C6)

Quantity	Value
White field mass, initial position	38
Black field mass, initial position	38
Attack identity	holds (both sides)
Equivariance violations, pawnless	0 / 0 (white/black)
Pawn anomaly (initial position)	$176 = 88 + 88$

Summary of what is proved

We proved three structural properties of the threat field: the attack identity (mass additivity), the exact D_4 -equivariance for orientation-neutral particles, and the finite pawn anomaly 176 for oriented ones. The mechanism is the interchange of summation, the geometric invariance of the attack relation, and a direct enumeration with the result frozen. The K3 layer now rests on a strict foundation: the threat field is an honest discrete potential with a known symmetry group.

Verification

Check: `python3 dynamics.py -run C6` or the C6 line of the polyglot core (the core verifies the pawnless equivariance and the anomaly 176; the attack identity — the laboratory). Equivariance is checked by comparing the fields of the original and transformed positions for all eight g ; PASS requires zero pawnless violations and exactly 176 in the initial position in all seven implementations.